Students performance Analysis

Turning Student Data into Actionable Insights

This presentation analyzes student academic performance using attendance, marks, study hours, assignments, and participation data. The goal is to identify trends, top performers, and improvement opportunities.

PPREPARED BY :-SHRADDHA WAGH
BSC DATA SCIENCE PROJECT

Project Overview

Understanding Student Performance Through Data

The Student Performance Analysis project focuses on evaluating academic results using key factors such as marks, attendance percentage, study hours, assignment completion, and class participation. By analyzing this data, we can understand student behavior, identify high achievers, and detect areas where students need support.

Process of Data Analysis

Student Data Analysis Process

- Collect Data – Gather student marks, attendance, study hours, assignments, and participation data.
- Clean Data – Remove errors, blanks, duplicates, and fix formatting issues.
- Analyze Data – Compare performance trends using formulas and charts.
- Interpret Results – Identify top performers and weak areas.
- Visualize Findings – Present results through KPI cards and dashboards.

1 COLLECT DATA

2 CLEAN DATA

3 ANALYZE DATA

4 INTERPRET RESULTS

5 VISUALIZE FINDINGS

Project Objectives

Goals of Student Performance Analysis

- Measure overall student academic performance.
- Analyze the impact of attendance on marks.
- Study relationship between study hours and results.
- Identify top-performing students.
- Detect low-performing students for improvement.
- Evaluate assignment completion and participation trends.
- Present insights using KPI cards and charts.

KPI Cards Page

total student
100

Avg Marks
69.3

Avg Attendance
81.02

- **KPI Card 1:**
- Total Students
- Shows total number of students in dataset
- Helps understand overall sample size
- Value: 100
- **KPI Card 2:**
- Average Attendance
- Displays average attendance percentage of all students
- Measures class regularity and discipline
- Value: 81.02%
- **KPI Card 3:**
- Average Marks
- Shows average marks scored by students
- Indicates overall academic performance level
- Value: 69.3

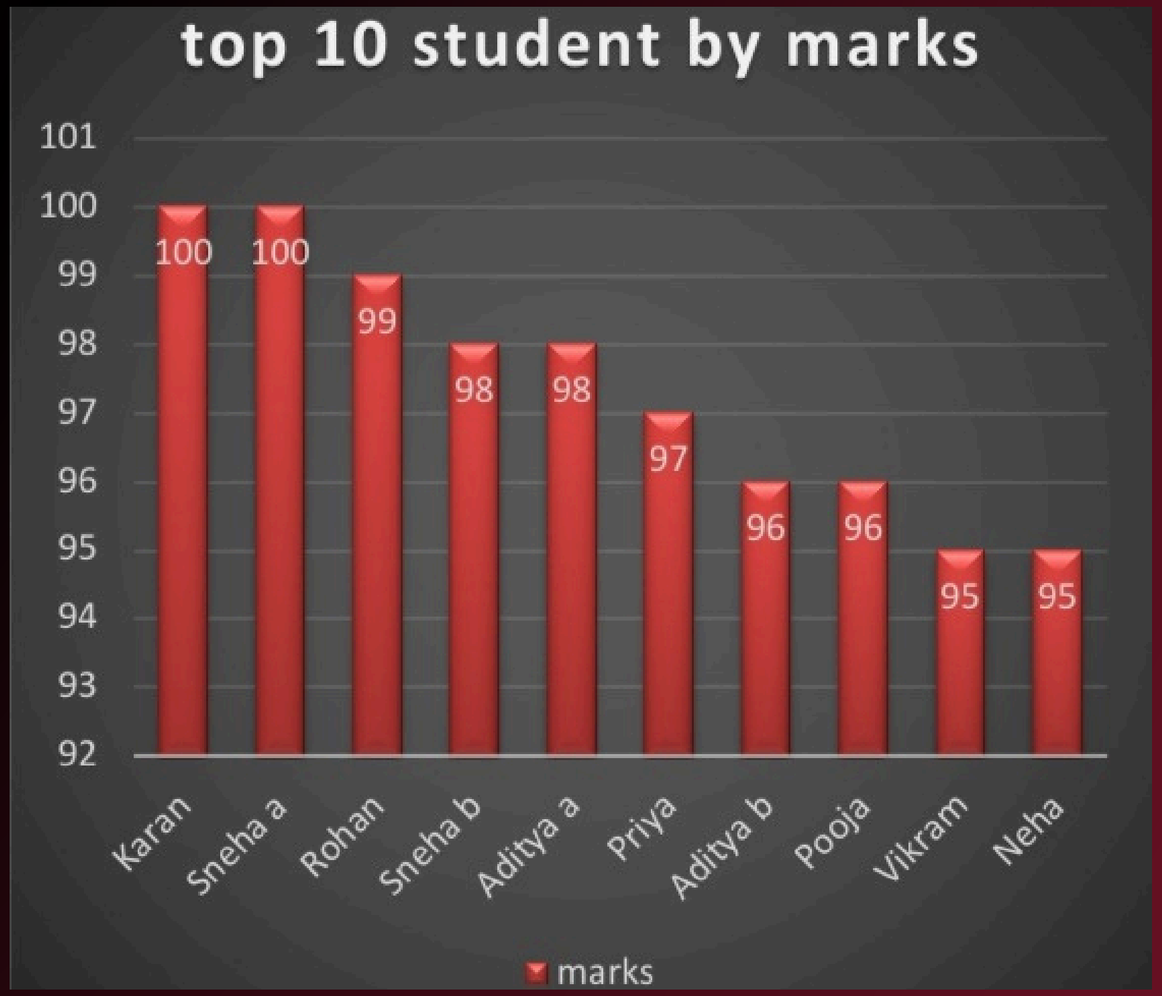
Top 10 Students Performance

Highest Scoring Students in the Dataset

This chart highlights the top 10 students based on marks obtained. It helps identify the best academic performers and compare score differences among high-achieving students.

key insights:-

- Karan and Sneha R scored the highest marks (100).
- Rachna secured second highest position (99).
- Performance gap among top students is very small.
- Most top students scored between 95 to 100.
- Consistent high scores indicate strong academic achievement.



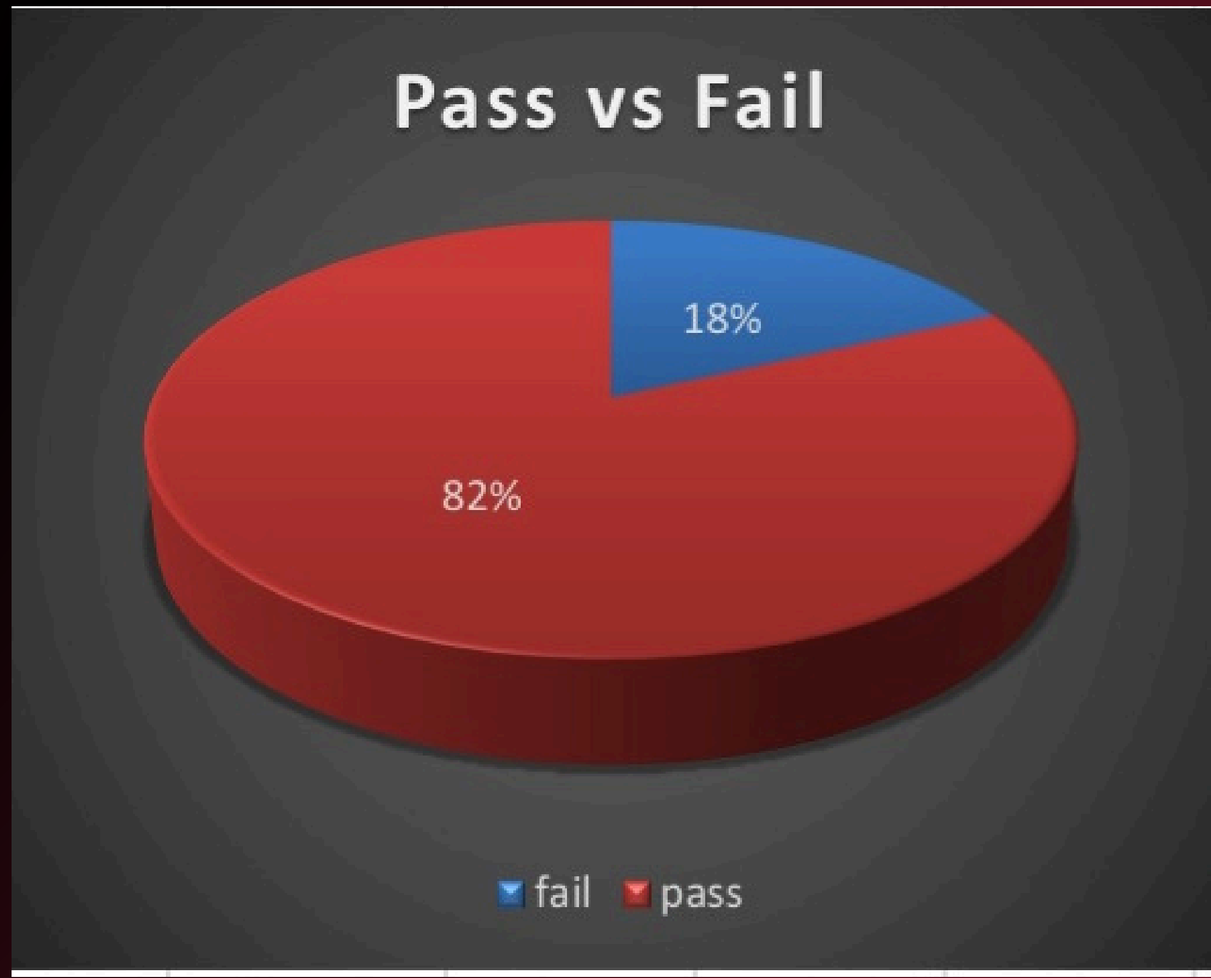
Pass vs Fail Analysis

Overall Result Distribution of Students

This chart shows the percentage of students who passed and failed in the dataset. It provides a quick view of the overall academic success rate.

Key Insights:

- 82% of students passed the examination.
- 18% of students failed.
- Majority of students achieved passing marks.
- Overall class performance is positive.
- Extra support can be given to failed students for improvement.



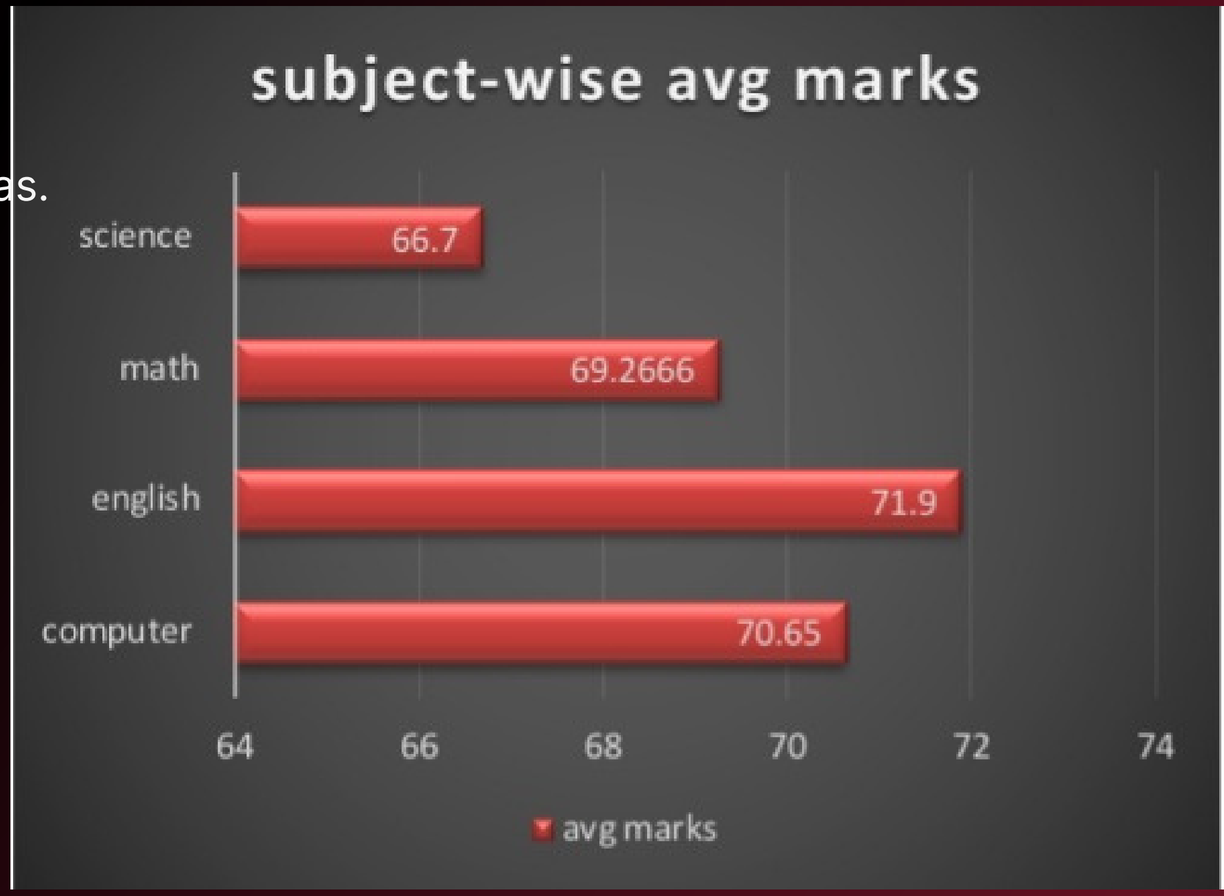
Subject-wise Average Marks

Comparing Performance Across Subjects

This chart presents the average marks scored by students in different subjects. It helps identify strong and weak subject areas.

Key Insights:

- English has the highest average marks.
- Computer subject also shows strong performance.
- Maths is in the middle range.
- Science has the lowest average marks.
- Subjects with lower averages may need additional focus and support.



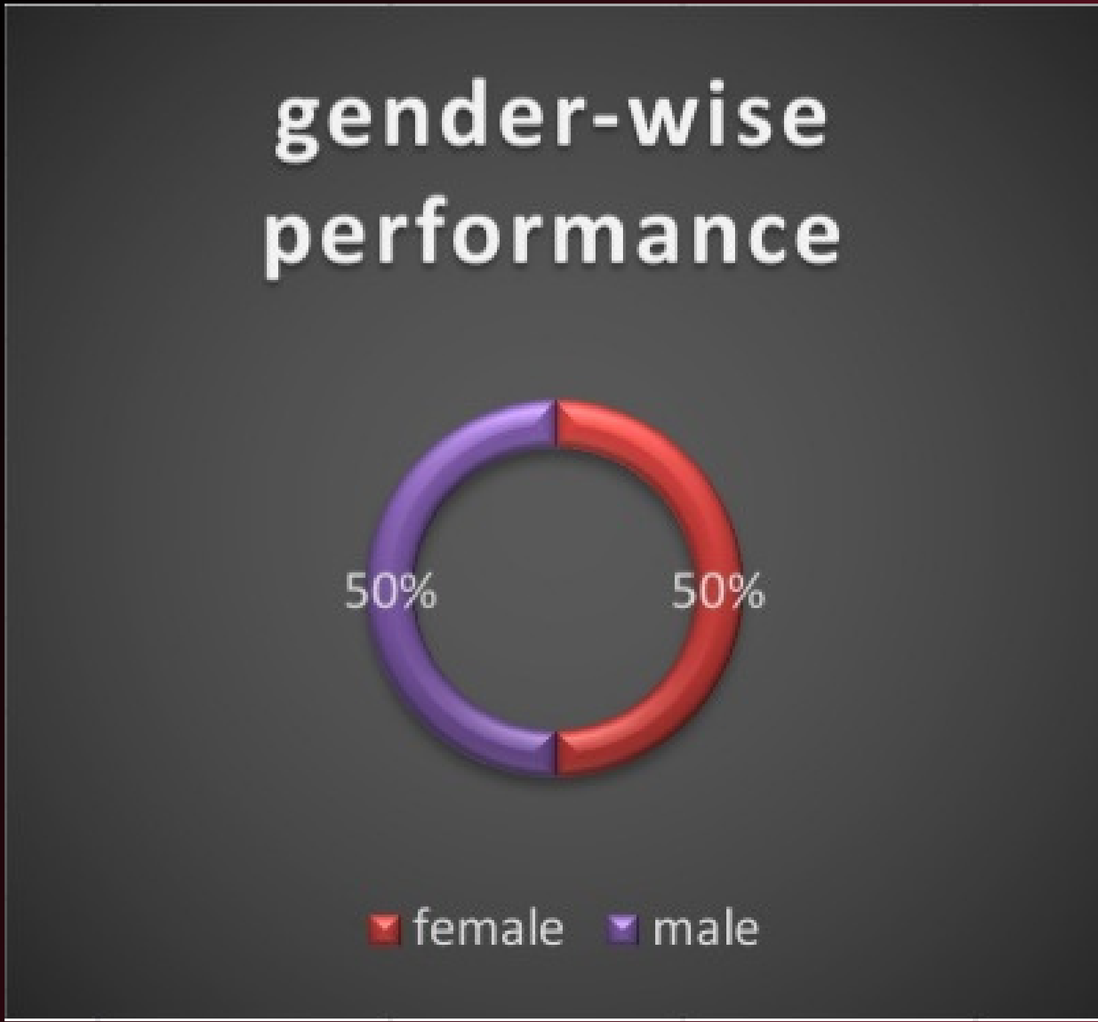
Gender-wise Performance

Student Distribution by Gender

This chart shows the gender-wise distribution of students i
n the dataset. It helps understand participation balance and
compare representation.

Key Insights:

- Male and female students are equally represented.
- Both groups contribute 50% each of total students.
- Dataset has balanced gender distribution.
- Equal representation supports fair performance analysis.
- Useful for future comparison of marks or attendance by gender.



Key Insights & Findings

Important Outcomes from Student Performance Analysis

Insights:

- Total of 100 students were analyzed in the dataset.
- Average attendance is 81.02%, showing good class regularity.
- Average marks are 69.3, indicating moderate overall performance.
- 82% students passed, while 18% need improvement support.
- Top students scored between 95 to 100 marks.
- English and Computer subjects performed better than others.
- Science subject needs more academic attention.
- Balanced gender distribution ensures fair analysis results.

Recommendations:

- Provide extra coaching for low-performing students.
- Focus more on weak subjects like Science.
- Encourage regular attendance and study habits.
- Reward top performers to motivate others.

Conclusion

Summary of Student Performance Analysis

The Student Performance Analysis project successfully transformed raw student data into meaningful insights using KPI cards and charts. The analysis highlighted overall academic performance, attendance trends, pass rates, top performers, and subject-wise strengths. These findings can help educators make better decisions, improve weak areas, and support student success through data-driven strategies.

Closing Points:

- Clear understanding of student performance
- Easy visualization through dashboard charts
- Supports better academic planning
- Helps improve future results

Thankyou

Thank You for Your Time & Attention

We appreciate your valuable time in reviewing this Student Performance Analysis presentation.
Your feedback and questions are welcome.

Questions?
Discussion Welcome

PREPARED BY: SHRADDHA WAGH
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